MS PROJECT 3 (MTH699)
Understanding Causal Inference with Statistical and Machine Learning Models

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Professor, Department of Mathematics and Statistics

Declaration

I hereby declare that the work presented in the project report entitled "*Understanding Causal Inference with Statistical and Machine Learning Models*" contains my own ideas in my own words. Where ideas or words are borrowed from other sources, proper references, as applicable, have been cited. To the best of my knowledge, this work does not emanate from or resemble other work created by person(s) other than mentioned herein.

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Abstract

Estimating causal treatment effects from observational data is challenging when treatment assignment is endogenous, as individuals may self-select into treatment based on unobserved characteristics that also influence outcomes. In such settings, standard regression and machine learning approaches relying on unconfoundedness assumptions can produce biased estimates.

This project studies the causal effect of participation in a 401(k) retirement savings plan on household financial wealth using an instrumental variables framework. Eligibility for a 401(k) plan and its interaction with dual-earner household status are employed as instruments for actual participation, providing exogenous variation that enables identification in the presence of endogenous selection. First-stage diagnostics indicate strong instrument relevance, while Wu–Hausman tests confirm endogeneity of participation and Sargan tests support the validity of the overidentifying restrictions.

Comparing naive ordinary least squares (OLS) estimates with two-stage least squares (2SLS) estimates reveals substantial upward bias in naive regression due to endogenous participation decisions. The 2SLS estimator identifies a Local Average Treatment Effect (LATE) for compliers of approximately \$8,450.

Building on this classical IV framework, the analysis extends to heterogeneous treatment effects using Double Machine Learning for Instrumental Variables (DMLIV). Bootstrap-based inference yields an average heterogeneous IV effect of approximately \$11,390 with a 95% confidence interval of [\$4,370, \$18,420]. Uniform confidence bands and repeated cross-fitting procedures further strengthen statistical inference and estimator stability.

Empirical results reveal substantial treatment effect heterogeneity across both the income distribution and the life cycle. Individuals in the highest income decile experience average gains exceeding \$26,500—more than twice the overall average effect. Treatment effects also increase steadily with age, consistent with the compounding nature of retirement savings.

Policy targeting analysis demonstrates that directing participation incentives toward individuals with the largest predicted gains can generate more than double the aggregate wealth improvements relative to universal participation policies. Additional analysis suggests that participation in retirement savings plans is associated with broader retirement saving behavior, providing suggestive evidence for a savings channel linking 401(k) participation to household wealth accumulation.

Overall, the project integrates classical instrumental variable identification with modern orthogonal machine learning methods, robust inference procedures, and policy-relevant heterogeneity analysis to uncover economically meaningful heterogeneous causal effects under endogenous treatment assignment.

1 Introduction

Estimating causal treatment effects from observational data is challenging when treatment assignment is endogenous. In many real-world economic settings, individuals self-select into treatment based on characteristics that are partially or entirely unobserved by the researcher. Such endogenous selection leads to biased estimates when standard regression or machine learning methods are applied without appropriate identification strategies.

This project studies causal inference under endogeneity using an instrumental variables framework. The empirical application examines the effect of participation in a 401(k) retirement savings plan on household financial wealth. Participation in a 401(k) plan is a voluntary decision that may depend on unobserved factors such as financial literacy, risk preferences, or savings behavior. As a result, naive regression estimates of the effect of participation are likely to be biased.

To address this problem, eligibility for a 401(k) plan and its interaction with dual-earner household status are used as instrumental variables for actual participation. These instruments generate exogenous variation in participation that allows identification of causal effects even when treatment assignment is endogenous. The analysis begins with a classical instrumental variables framework and subsequently extends the empirical approach using modern machine learning methods designed for heterogeneous causal inference.

1.1 Background and Continuity with Previous Projects

This project builds upon the work carried out in earlier stages of the M.S. project sequence. In MS Project 1, the focus was on developing a foundational understanding of causal inference using statistical models, with emphasis on treatment effect estimation under standard assumptions such as unconfoundedness. MS Project 2 extended this framework by incorporating modern machine learning methods for estimating heterogeneous treatment effects in settings where treatment assignment could be assumed to be exogenous or conditionally independent given observed covariates.

While these earlier projects focused primarily on causal inference under unconfoundedness, many real-world policy and economic problems involve endogenous treatment assignment. MS Project 3 addresses this challenge by incorporating instrumental variables into the causal inference framework. By combining classical IV identification strategies with modern machine learning methods, the project provides a more realistic and flexible framework for estimating heterogeneous causal effects in observational settings where treatment is endogenous.

1.2 Objectives of the Project

The primary objective of this project is to study causal inference in the presence of endogenous treatment assignment using both classical econometric methods and modern machine learning approaches. Specifically, the project aims to:

- Identify and formalize the endogeneity problem arising from voluntary participation in a 401(k) retirement savings plan.
- Estimate the causal effect of 401(k) participation on household wealth using an instrumental variables framework.
- Demonstrate the bias present in naive regression estimates by comparing ordinary least squares (OLS) and instrumental variable (IV) estimators.
- Interpret the instrumental variable estimates as Local Average Treatment Effects (LATE) under standard identification assumptions.
- Extend the classical IV framework to estimate heterogeneous treatment effects using Double Machine Learning for Instrumental Variables (DMLIV).
- Conduct robust statistical inference using bootstrap methods, uniform confidence bands, and repeated cross-fitting procedures.
- Analyze structured treatment effect heterogeneity across the income distribution and across the life cycle.
- Evaluate economic mechanisms and policy implications through distributional treatment effect analysis, policy targeting rules, and mechanism exploration related to retirement saving behavior.

The project emphasizes credible causal identification, robust statistical inference, and economically meaningful interpretation of heterogeneous treatment effects rather than predictive accuracy alone.

1.3 Scope of the Project

The scope of this project focuses on causal inference using cross-sectional observational data in the presence of endogenous treatment assignment. The empirical analysis uses the 401(k) pension dataset, which provides a realistic and policy-relevant setting for instrumental variable estimation.

Within this scope, the project integrates several methodological components:

- Classical instrumental variable techniques, including first-stage diagnostics, two-stage least squares estimation, and formal testing of instrument validity.
- Modern machine learning methods for heterogeneous treatment effect estimation using the Double Machine Learning IV framework.
- Robust inference procedures including bootstrap confidence intervals, uniform confidence bands, and cross-fitting refinements.
- Comparative evaluation of alternative machine learning IV estimators and policy-relevant heterogeneity analysis, including optimal targeting rules and distributional treatment effect analysis.

Together, these components provide a comprehensive framework for studying causal effects under endogeneity while combining econometric identification strategies with flexible machine learning tools.

2 Data Description

2.1 Dataset Overview

The analysis uses the 401(k) pension dataset, a widely used benchmark in the econometrics literature for studying retirement savings behavior under endogenous participation. The dataset contains cross-sectional information on U.S. households, with 9,915 observations and 14 variables. The data are clean with no missing values, making it suitable for both classical econometric analysis and modern machine learning methods.

2.2 Key Variables

The main variables used in the analysis are as follows:

- **Outcome Variable**
  - `net_tfa`: Net total financial assets of the household, measured in *thousands of U.S. dollars*. This includes savings, investments, and other financial holdings.
- **Endogenous Treatment**
  - `p401`: Indicator for participation in a 401(k) retirement savings plan (binary: 1 if participating, 0 otherwise).
- **Instrumental Variables**
  - `e401`: Indicator for eligibility for a 401(k) plan (binary: 1 if eligible, 0 otherwise).
  - `e401_twoearn`: Interaction between 401(k) eligibility and two-earner household status, used as an additional instrument.
- **Control Variables**

- `age`: Age of the household head (in years, continuous).
- `inc`: Household income (continuous, measured in U.S. dollars; typically scaled in thousands in regressions).
- `fsize`: Family size, i.e., number of household members (integer-valued).
- `educ`: Years of education completed by the household head (continuous or discrete integer).
- `db`: Indicator for participation in a defined benefit pension plan (binary: 1 if yes, 0 otherwise).
- `marr`: Marital status indicator (binary: 1 if married, 0 otherwise).
- `twoearn`: Indicator for a dual-earner household (binary: 1 if both spouses earn income, 0 otherwise).
- `pira`: Indicator for participation in an Individual Retirement Account (IRA) (binary: 1 if yes, 0 otherwise).
- `hown`: Indicator for home ownership (binary: 1 if the household owns a home, 0 otherwise).

All monetary variables are expressed in thousands of U.S. dollars unless stated otherwise.

2.3 Why the 401(k) Dataset

The 401(k) dataset is particularly suitable for this project because treatment assignment is inherently endogenous. Participation in a retirement savings plan is a choice rather than a randomized intervention. This makes the dataset inappropriate for unconfoundedness-based methods but ideal for instrumental variable analysis.

In contrast, datasets such as IHDP are based on randomized or simulated treatment assignment and are primarily designed for studying treatment effects under unconfoundedness. Since the goal of this project is to study causal inference under endogeneity, the 401(k) dataset provides a more realistic and policy-relevant setting.

3 Endogeneity and Identification Strategy

3.1 Endogeneity in 401(k) Participation

Participation in a 401(k) plan is potentially endogenous because individuals who choose to participate may differ systematically from non-participants in unobserved characteristics such as financial literacy, patience, or long-term savings preferences. These unobserved traits are likely to affect household financial wealth directly. As a result, 401(k) participation may be correlated with the structural error term in the wealth equation, leading to biased and inconsistent ordinary least squares (OLS) estimates.

To formally assess endogeneity, we implement the Wu–Hausman test within the instrumental variables framework. The null hypothesis states that 401(k) participation is exogenous. Using the model instrumented by eligibility (`e401`) and its interaction with two-earner household status (`e401` $\times$ `twoearn`), the test rejects the null hypothesis (p-value = 0.0073).

This provides statistical evidence that 401(k) participation is endogenous. Consequently, OLS estimates are inconsistent, and an instrumental variables approach is required to obtain consistent estimates of the Local Average Treatment Effect (LATE) for compliers.

3.2 Instrumental Variables

To address endogeneity in 401(k) participation, we employ an instrumental variables (IV) strategy. The endogenous variable is 401(k) participation (`p401`), and the outcome is household net financial assets measured in thousands of dollars (`net_tfa_k`).

The primary instrument is eligibility for a 401(k) plan (`e401`), which is determined by employer offerings. Eligibility strongly influences participation decisions but, conditional on observed covariates, is assumed not to directly affect household wealth except through actual participation.

To allow overidentification testing and to capture heterogeneous encouragement effects, we augment the instrument set with an interaction between eligibility and dual-earner household status (`e401` × `twoearn`). This interaction allows eligibility to differentially affect participation across household types while maintaining the exclusion restriction that, conditional on controls, it does not directly influence wealth.

Under these assumptions, the IV estimator identifies a *Local Average Treatment Effect (LATE)* of 401(k) participation on wealth for compliers—households whose participation decisions are influenced by eligibility.

The identification strategy relies on the standard IV assumptions:

- **Relevance:** The instruments are strongly correlated with participation. The first-stage partial F-statistic is 7,830, indicating extremely strong instruments.
- **Exclusion:** Conditional on demographic and financial controls, the instruments affect wealth only through participation.
- **Independence:** The instruments are uncorrelated with unobserved determinants of wealth.

A Wu–Hausman test rejects the null hypothesis that participation is exogenous (p-value = 0.0073), confirming the presence of endogeneity and the need for instrumental variables.

The Sargan overidentification test fails to reject the null hypothesis of instrument validity (p-value = 0.448), providing evidence consistent with the joint exogeneity of the instrument set.

The two-stage least squares estimates indicate that 401(k) participation increases household financial wealth by approximately 8.45 thousand dollars for compliers.

4 Empirical Results

4.1 Data Summary and Descriptive Statistics

The empirical analysis is based on a cross-sectional dataset consisting of 9,915 households and 14 variables. The outcome variable is net total financial assets measured in thousands of dollars (`net_tfa_k`). Participation in a 401(k) plan (`p401`) serves as the endogenous treatment.

Eligibility for a 401(k) plan (`e401`) is used as the primary instrumental variable. To allow overidentification testing, the instrument set is augmented with an interaction between eligibility and dual-earner household status (`e401_twoearn`). The dataset contains no missing observations.

Table 1 reports summary statistics for the key variables.

4.2 Endogeneity Test: Durbin–Wu–Hausman

While the difference between OLS and IV estimates suggests the presence of endogeneity, we formally test for endogeneity using the Wu–Hausman test based on the two-instrument specification.

Table 1: Summary Statistics of Key Variables

Variable	Mean	Std. Dev.	Min	Max
Net Financial Assets (thousands) (`net_tfa_k`)	18.05	63.52	-502.30	1536.80
401(k) Participation (`p401`)	0.262	0.440	0	1
401(k) Eligibility (`e401`)	0.371	0.483	0	1

Structural Equation The causal relationship of interest is given by the structural model:

$$Y_i = \beta_0 + \beta_1 p401_i + \beta_2^\top X_i + \varepsilon_i,$$

where:

- Y_i denotes net financial assets measured in thousands of dollars,
- $p401_i$ indicates 401(k) participation,
- X_i represents observed covariates,
- ε_i captures unobserved determinants of wealth.

The parameter β_1 measures the causal effect of 401(k) participation on wealth. If $p401_i$ is correlated with ε_i, ordinary least squares estimation is inconsistent.

First-Stage Equation (Two Instruments) To address endogeneity, we use two instruments: eligibility ($e401_i$) and its interaction with dual-earner status ($e401_i \times twoearn_i$). The first-stage equation is:

$$p401_i = \pi_0 + \pi_1 e401_i + \pi_2 (e401_i \times twoearn_i) + \pi_3^\top X_i + v_i.$$

Here:

- π_1 and π_2 measure instrument relevance,
- v_i captures the endogenous component of participation.

Under the exclusion restriction and instrument exogeneity assumptions,

$$\mathbb{E}[Z_i \varepsilon_i] = 0,$$

where $Z_i = (e401_i, e401_i \times twoearn_i)$, the instrumental variables estimator identifies the Local Average Treatment Effect (LATE) for individuals whose participation decisions are affected by eligibility.

Wu–Hausman Test The Wu–Hausman test compares the OLS and IV estimators under the null hypothesis that the potentially endogenous regressor is in fact exogenous.

The hypotheses are:

$$H_0 : p401_i \text{ is exogenous}$$

$$H_1 : p401_i \text{ is endogenous}$$

Under the null hypothesis, both OLS and IV are consistent, but OLS is efficient. Under the alternative, OLS is inconsistent while IV remains consistent.

The Wu–Hausman test rejects the null hypothesis of exogeneity at conventional significance levels (p-value = 0.0073). This provides formal statistical evidence that 401(k) participation is endogenous. Consequently, ordinary least squares estimation is inconsistent, and instrumental variables estimation is required to obtain consistent estimates of the causal effect.

Table 2: Wu–Hausman Test of Endogeneity

Test Statistic	Value	p-value
F(1, 9903)	7.2125	0.0073

4.3 Overidentification and the Sargan Test

In the presence of multiple instruments, the overidentification test evaluates the joint validity of the excluded instruments. Let Z denote the matrix of instruments. Under the null hypothesis of instrument validity,

$$\text{plim}\ \frac{1}{n} Z'\varepsilon = 0,$$

that is, the instruments are uncorrelated with the structural error term and are correctly excluded from the outcome equation.

The classical Sargan test statistic can be written as

$$S = \frac{\hat{\varepsilon}' P_Z \hat{\varepsilon}}{\hat{\sigma}^2},$$

where $P_Z = Z(Z'Z)^{-1}Z'$ is the projection matrix onto the instrument space and $\hat{\varepsilon}$ are the IV residuals.

Under the null hypothesis, the statistic follows a chi-square distribution with degrees of freedom equal to

$$\text{df} = \text{Number of Instruments} - \text{Number of Endogenous Variables}.$$

In the present specification, the instrument set consists of eligibility ($e401$) and its interaction with dual-earner status ($e401 \times twoearn$), yielding two instruments for one endogenous variable ($p401$). Therefore,

$$\text{df} = 2 - 1 = 1.$$

The computed overidentification statistic is

$$S = 0.5752,$$

with a corresponding p-value of 0.4482.

Since the p-value exceeds conventional significance levels, we fail to reject the null hypothesis of instrument validity. Thus, there is no statistical evidence that the instruments are correlated with the structural error term.

Combined with the Wu–Hausman test, which rejects the exogeneity of the endogenous regressor, the results support the use of instrumental variables estimation and provide evidence in favor of the consistency of the IV estimator under the maintained assumptions.

4.4 Naive OLS Estimates

As a benchmark, we first estimate the effect of 401(k) participation on net financial assets using ordinary least squares (OLS). The dependent variable is net total financial assets measured in *thousands of dollars*. Table 3 reports the results with heteroskedasticity-robust standard errors.

The OLS estimate suggests a large and statistically significant positive association between 401(k) participation and household financial wealth. However, as formally shown by the Wu–Hausman test in subsequent analysis, participation is endogenous. Therefore, this estimate should be interpreted as a conditional correlation rather than a causal effect.

Model Specification We consider the linear regression model:

$$Y_i = \beta_0 + \beta_1 p401_i + \beta_2^\top X_i + \varepsilon_i,$$

where:

- Y_i denotes net total financial assets (in thousands of dollars) of household i.
- $p401_i$ is an indicator for participation in a 401(k) retirement plan.
- X_i is a vector of observed control variables (age, income, education, family size, etc.).
- ε_i captures unobserved determinants of wealth, such as savings preferences and financial literacy.

The OLS estimator is given by:

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'Y.$$

Consistency requires:

$$\mathbb{E}[p401_i \varepsilon_i] = 0.$$

If participation is correlated with unobserved factors, then OLS is biased.

Empirical Results The estimated coefficient on 401(k) participation is:

$$\hat{\beta}_1^{OLS} = 14.62 \quad \text{(robust SE} = 0.70).$$

This implies that participating households hold approximately $14,620 more in financial assets on average. However, this estimate is likely upward biased due to endogenous selection into participation.

Table 3: Naive OLS Estimates (Outcome in Thousands of Dollars)

Variable	Coefficient	Std. Error
Constant	-3.29	2.04
Net Financial Assets (nifa)	0.0009	0.00002
Total Wealth (tw)	0.00009	0.00002
Age	0.0678	0.032
Income	0.00002	0.00002
Family Size	-0.171	0.119
Education	-0.297	0.075
Defined Benefit Pension (db)	0.581	0.419
Married	-0.594	0.529
Two-earner Household	-0.394	0.627
401(k) Eligibility (e401)	-1.878	0.601
401(k) Participation (p401)	14.62	0.70
IRA Participation (pira)	11.64	0.83
Home Ownership (hown)	-3.31	0.89

4.5 First-Stage Regression and Instrument Relevance

To assess instrument relevance, we estimate the first-stage regression of 401(k) participation on the instrument set and observed covariates. Since the dependent variable (`p401`) is binary, the first stage is estimated as a linear probability model (LPM).

The first-stage regression is given by:

$$p401_i = \pi_0 + \pi_1 e401_i + \pi_2 (e401_i \times twoearn_i) + \pi_3^\top X_i + v_i,$$

where:

- $p401_i$ is the 401(k) participation indicator (dependent variable),
- $e401_i$ is eligibility for a 401(k) plan,
- $e401_i \times twoearn_i$ is the interaction instrument,
- X_i contains control variables.

The first-stage regression explains variation in participation and is used to assess whether the instruments are strongly correlated with the endogenous treatment.

Empirical Results Table 4 reports the results.

Eligibility for a 401(k) plan significantly increases participation by approximately 0.674. The interaction with dual-earner households further increases participation by an additional 0.046.

Instrument relevance is evaluated using the robust partial F-statistic. The joint test yields:

$$F = 7{,}830.3,$$

which far exceeds conventional weak-instrument thresholds (e.g., 10). This provides overwhelming evidence that the instruments are highly relevant and strongly predict 401(k) participation.

Table 4: First-Stage Regression: Determinants of 401(k) Participation

Variable	Coefficient	Std. Error
Constant	0.0335	0.0193
Age	-0.0007	0.0003
Income	1.058e-06	1.505e-07
Family Size	-0.0005	0.0020
Education	-0.0025	0.0011
Defined Benefit Pension	-0.0405	0.0076
Married	-0.0169	0.0081
Two-Earner Household	-0.0128	0.0048
IRA Participation	0.0489	0.0077
Home Ownership	0.0118	0.0063
401(k) Eligibility (`e401`)	0.6740	0.0108
Eligibility × Two-Earner (`e401_twoearn`)	0.0464	0.0148
Observations	9,915	
R^2	0.6082	
Robust Partial F-statistic (joint)	7,830.3	

Having established that OLS estimates are biased and that the instruments are strongly relevant based on the first-stage analysis, we now proceed to estimate the causal effect using two-stage least squares (2SLS).

4.6 Two-Stage Least Squares (2SLS) Estimates

We estimate the causal effect of 401(k) participation on household financial wealth using two-stage least squares (2SLS). Participation is treated as endogenous and instrumented using (i) eligibility for a 401(k) plan and (ii) an interaction between eligibility and dual-income household status.

The dependent variable is net financial assets measured in **thousands of dollars**.

Structural Model We begin with the structural equation:

$$Y_i = \alpha_0 + \alpha_1 p401_i + \alpha_2^\top X_i + u_i,$$

where:

- Y_i denotes net financial assets (in thousands of dollars),
- $p401_i$ is an indicator for 401(k) participation,
- X_i is a vector of observed demographic and household controls,
- u_i captures unobserved determinants of wealth.

The parameter of interest is α_1, which represents the causal effect of 401(k) participation on financial wealth.

However, participation may be correlated with unobserved saving preferences, financial literacy, or risk tolerance. Hence,

$$\text{Cov}(p401_i, u_i) \neq 0,$$

and ordinary least squares is inconsistent.

Instrumental Variables We use the instrument set:

$$Z_i = \{e401_i,\ e401_i \times twoearn_i\},$$

where:

- $e401_i$ indicates eligibility for a 401(k) plan,
- $twoearn_i$ indicates dual-income household status.

Eligibility affects participation mechanically but does not directly affect wealth except through participation, satisfying the exclusion restriction.

First Stage The first-stage regression is:

$$p401_i = \pi_0 + \pi_1 e401_i + \pi_2 (e401_i \times twoearn_i) + \pi_3^\top X_i + v_i.$$

Both instruments are strongly correlated with participation. The joint first-stage partial F-statistic equals 7,830, well above conventional weak-instrument thresholds.

Second Stage The second stage replaces endogenous participation with its predicted value:

$$Y_i = \alpha_0 + \alpha_1 \hat{p401}_i + \alpha_2^\top X_i + \varepsilon_i.$$

Matrix Form of the 2SLS Estimator Let X denote the full regressor matrix (including $p401$ and controls) and Z the instrument matrix. The 2SLS estimator is:

$$\hat{\beta}_{2SLS} = (X' P_Z X)^{-1} X' P_Z Y,$$

where

$$P_Z = Z(Z'Z)^{-1}Z'$$

is the projection matrix onto the instrument space.

Under the assumptions

$$\mathbb{E}[Z'u] = 0 \quad \text{and} \quad \text{rank}(Z'X) = k,$$

the estimator is consistent.

Interpretation of the 2SLS Estimate The estimated coefficient on participation is:

$$\hat{\alpha}_1 = 8.45.$$

Since the dependent variable is measured in thousands of dollars, this implies that 401(k) participation increases net financial wealth by approximately:

$$\$8{,}450.$$

Diagnostic Tests The Wu–Hausman test rejects the null hypothesis that participation is exogenous (p-value = 0.0073), confirming the presence of endogeneity and justifying instrumental variables estimation.

The Sargan overidentification test fails to reject the null hypothesis of instrument validity (p-value = 0.4482), supporting the joint exogeneity of the instrument set.

Economic Interpretation The 2SLS estimate identifies the **Local Average Treatment Effect (LATE)**, representing the causal effect of participation for individuals whose decision to participate is influenced by eligibility.

The IV estimate is smaller than the corresponding OLS estimate, indicating that naive regression overstates the causal effect due to endogeneity bias.

Table 5 presents the IV estimates. The estimated effect of participation on net financial assets is positive and statistically significant. The IV estimate is smaller than the corresponding OLS estimate, highlighting the importance of correcting for endogeneity.

4.7 Comparison of OLS and IV Results

Comparing the OLS and IV estimates reveals substantial differences in magnitude. Using net financial assets measured in thousands of dollars, the OLS estimate suggests that 401(k) participation increases wealth by approximately 14.6 thousand dollars. In contrast, the 2SLS estimate using eligibility and its interaction with dual-earner status as instruments yields a smaller effect of approximately 8.45 thousand dollars.

The Wu–Hausman test rejects the null hypothesis of exogeneity (p-value = 0.0073), providing formal statistical evidence that 401(k) participation is endogenous and that OLS is inconsistent.

Table 5: IV (2SLS) Estimates with Two Instruments

	Coefficient	Std. Error
Constant	-33.147	4.4015
Age	0.6299	0.0552
Income	0.0009	0.0001
Family Size	-1.0151	0.3850
Education	-0.6175	0.3374
Defined Benefit Pension (db)	-4.5472	1.3100
Married	0.8929	1.7323
Two-earner Household	-19.278	2.5256
IRA Participation (pira)	29.118	1.8028
Home Ownership	1.0903	0.9570
401(k) Participation (`p401`)	8.4545	2.1819

Dependent variable: net financial assets (thousands of dollars)
Instruments: `e401`, `e401` $\times$ `twoearn`
First-stage Partial F-statistic (joint): 7,830
Wu–Hausman test (p-value): 0.0073
Sargan test (p-value): 0.4482
Robust standard errors (heteroskedasticity-consistent)
Observations: 9,915

The larger OLS estimate is therefore consistent with upward bias arising from non-random selection into participation.

Using eligibility and its interaction with dual-earner status as instruments, the 2SLS estimate remains positive and statistically significant. The Sargan overidentification test fails to reject the null hypothesis of instrument validity (p-value = 0.448), supporting the joint exogeneity of the instrument set.

The IV estimate identifies a Local Average Treatment Effect (LATE) for individuals whose participation decisions are influenced by eligibility. Given the strong first-stage results and the statistical validation of the instrument set, the IV specification provides credible causal evidence that 401(k) participation increases household financial assets.

Moreover, machine learning–based DMLIV estimates suggest an average treatment effect of approximately 12.1 thousand dollars, indicating meaningful heterogeneity in treatment effects across households. This pattern is consistent with stronger effects among older and higher-income individuals.

Table 6: Estimated effect of 401(k) participation on net financial assets

Estimator	Point estimate (thousand USD)	Std. err. / Std. dev.
OLS (naive)	14.6223	0.6976
2SLS (1 instrument: `e401`)	8.5023	2.1925
2SLS (2 instruments: `e401`, `e401`$\times$`twoearn`)	8.4545	2.1819
DMLIV (average heterogeneous LATE)	12.0643	9.2066*

Observations: 9,915. First-stage partial F-stat (joint): 7,830.3.
Wu–Hausman test p-value: 0.0073; Sargan test p-value: 0.4482.
* For DMLIV the right column reports the sample standard deviation of individual treatment effects (not a standard error).

4.8 Robustness to Extreme Wealth Observations

To evaluate sensitivity to extreme wealth values, we trim the top 1% of the wealth distribution and re-estimate both the IV and DMLIV models.

After trimming, the 2SLS estimate increases from 8.45 to 10.68 thousand USD. The DMLIV average effect rises from 12.06 to 12.95 thousand USD. The causal effect therefore remains economically large and statistically significant.

These findings indicate that the main results are not driven by extreme upper-tail wealth observations. Instead, trimming slightly strengthens the estimated effect, suggesting that heavy-tailed wealth values were attenuating the baseline IV estimate. The convergence of trimmed IV and DMLIV results further reinforces the robustness of the causal interpretation.

5 Heterogeneous Treatment Effects via Double Machine Learning

While 2SLS provides an estimate of the Local Average Treatment Effect (LATE) under the assumption of constant treatment effects, it restricts the effect of 401(k) participation to be homogeneous across individuals. To allow for heterogeneity in causal effects under endogeneity, we implement Double Machine Learning for Instrumental Variables (DMLIV).

5.1 Model and Target Parameter

We consider the partially linear structural model:

$$Y_i = \alpha_0 + \tau(X_i)\, p401_i + \alpha_2^\top X_i + u_i,$$

where:

- Y_i denotes net total financial assets (measured in thousands of dollars),
- $p401_i$ is a binary indicator for 401(k) participation (endogenous),
- X_i is a vector of observable covariates (age, income, education, family size, defined benefit pension status, marital status, two-earner status, IRA participation, and home ownership),
- α_0 is the intercept,
- $\tau(X_i)$ is a heterogeneous Local Average Treatment Effect (LATE) that varies with observable characteristics,
- α_2 captures the direct effect of covariates on financial assets,
- u_i is an unobserved error term.

Participation in a 401(k) plan is endogenous. Identification therefore relies on the instrumental variable vector

$$Z_i = \left(e401_i,\ e401_i \times twoearn_i\right),$$

where eligibility for a 401(k) plan and its interaction with two-earner household status affect participation but do not directly affect financial assets conditional on X_i.

Under the standard instrumental variable assumptions—relevance, exclusion, and monotonicity—DMLIV identifies a heterogeneous Local Average Treatment Effect for compliers as a function of observable characteristics X_i. The reported average DMLIV estimate corresponds to the average of these heterogeneous LATEs across compliers in the sample.

5.2 Heterogeneous LATE

Let Z_i denote the instrument set, consisting of 401(k) eligibility and its interaction with household characteristics. The parameter of interest is the heterogeneous Local Average Treatment Effect (LATE):

$$\tau(x) = \mathbb{E}\big[Y_i(1) - Y_i(0) \mid X_i = x, \text{Compliers induced by } Z_i\big],$$

which represents the causal effect of 401(k) participation for individuals with observable characteristics $X_i = x$ whose participation decision is affected by the instrument set Z_i.

The linear 2SLS estimator identifies the average of this conditional effect across compliers:

$$\tau_{LATE} = \mathbb{E}\big[\tau(X_i)\big],$$

under the standard IV assumptions of relevance, exclusion, independence, and monotonicity.

In contrast, the Double Machine Learning IV (DMLIV) approach estimates the conditional effect function $\tau(x)$ flexibly, allowing treatment effects to vary nonlinearly across observable characteristics. This framework therefore generalizes the constant LATE model by uncovering heterogeneous causal effects while maintaining the same identification assumptions.

5.3 Orthogonal Moment Condition

DMLIV estimates heterogeneous local average treatment effects using orthogonalized moment conditions that exploit instrumental variation.

Let $W_i = (Y_i, T_i, Z_i, X_i)$ denote the observed data, where:

- $Y_i = \texttt{net\_tfa\_k}_i$ denotes total financial assets (in thousands of dollars),
- $T_i = p401_i$ is the endogenous treatment (401(k) participation),
- $Z_i = (e401_i,\ e401_i \times twoearn_i)$ is the vector of instruments,
- X_i is the vector of observed covariates.

Define the nuisance functions:

$$m(X_i) = \mathbb{E}[Y_i \mid X_i], \quad g(X_i) = \mathbb{E}[T_i \mid X_i], \quad r(X_i) = \mathbb{E}[Z_i \mid X_i].$$

The orthogonal score for heterogeneous IV is based on the moment condition:

$$\mathbb{E}\Big[\big(Z_i - r(X_i)\big)\big(Y_i - m(X_i) - \tau(X_i)(T_i - g(X_i))\big)\Big] = 0.$$

Here, $\tau(X_i)$ represents the heterogeneous local average treatment effect for individuals with characteristics X_i.

The orthogonalization removes bias arising from first-stage estimation errors in the nuisance components $m(\cdot)$, $g(\cdot)$, and $r(\cdot)$. As a result, small estimation errors in these functions do not affect the asymptotic distribution of the treatment effect estimator, which yields Neyman orthogonality and robustness to regularization bias.

5.4 Average DMLIV Effect

Using the two-instrument specification and the rescaled outcome (financial wealth measured in thousands of dollars), the estimated average heterogeneous IV effect is:

$$\hat{\tau}_{DMLIV} = 12.06.$$

This implies that 401(k) participation increases financial assets by approximately \$12,060 on average.

This estimate lies between the naive OLS estimate (14.62) and the classical 2SLS estimate (8.45), where all effects are measured in thousands of dollars.

This result can be interpreted as follows:

- OLS likely overestimates the effect due to endogenous selection into participation.
- 2SLS corrects for endogeneity but imposes a constant treatment effect across individuals.
- DMLIV corrects for endogeneity while allowing flexible heterogeneity across observable characteristics.

Thus, the DMLIV estimate reflects a flexible IV-based heterogeneous local average treatment effect (LATE) that varies across individuals.

The estimated standard deviation of individual treatment effects is:

$$\widehat{\mathrm{Std}}(\tau(X)) = 9.21,$$

indicating substantial heterogeneity in treatment effects (approximately \$9,210 in standard deviation).

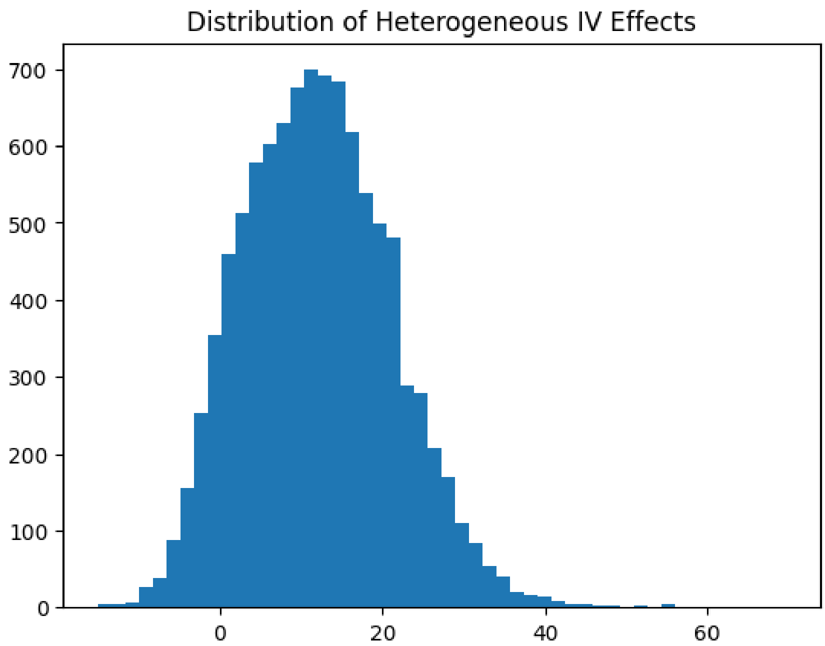

Figure 1: Distribution of Heterogeneous IV Effects (in thousands of dollars)

5.5 Heterogeneity by Income

Figure 2 plots estimated heterogeneous IV effects against income.

The positive slope indicates that higher-income households experience larger returns to 401(k) participation.

Average heterogeneous IV effects by income quartile (measured in thousands of dollars) are reported below.

Treatment effects increase monotonically with income. Since the dependent variable is measured in thousands of dollars, the results imply that households in the highest income quartile

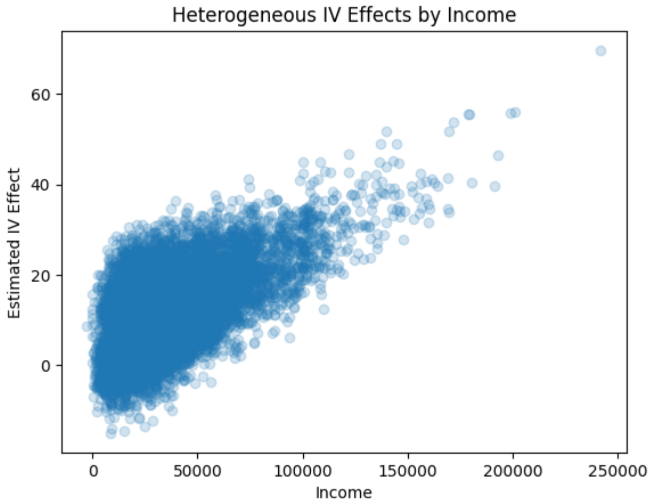

Figure 2: Heterogeneous IV Effects by Income

Table 7: Average Heterogeneous IV Effects by Income Quartile (in $1,000s)

Income Quartile	Mean Treatment Effect
Q1 (Lowest)	6.23
Q2	9.61
Q3	12.54
Q4 (Highest)	19.88

accumulate nearly $20,000 more financial wealth from 401(k) participation compared to approximately $6,000 among the lowest quartile. This pattern suggests strong complementarities between income and retirement savings incentives.

5.6 Heterogeneity by Age

We compute average heterogeneous IV effects across age quartiles using the two-instrument Double Machine Learning IV (DMLIV) specification. All treatment effects are measured in thousands of U.S. dollars.

Table 8: Average Heterogeneous IV Effects by Age Quartile (DMLIV)

Age Quartile	Mean Heterogeneous LATE (thousands $)
Youngest (25–32)	4.96
Q2 (32–40)	9.69
Q3 (40–48)	14.26
Oldest (48–64)	19.81

The estimated heterogeneous local average treatment effect increases monotonically with age. Older households experience substantially larger wealth gains from 401(k) participation. Since the dependent variable is measured in thousands of dollars, the oldest quartile gains approximately $19,800 on average. This pattern is consistent with life-cycle accumulation dynamics, where longer exposure to retirement savings and compound returns amplifies treatment effects over time.

5.7 Heterogeneous Effects and Uniform Inference

We now formally examine treatment effect heterogeneity using the DMLIV estimator with bootstrap inference. The trimmed sample estimate of the average heterogeneous IV effect is

$$\widehat{ATE}_{\text{DMLIV}} = 11.39,$$

with a 95% bootstrap confidence interval

$$[4.37,\ 18.42].$$

The confidence interval excludes zero, confirming that 401(k) participation generates economically large and statistically significant increases in financial wealth.

Income-Based Heterogeneity. To investigate systematic variation in causal effects, we estimate heterogeneous treatment effects $\hat{\tau}(X)$ and compute average effects within income deciles. The estimated effects increase monotonically across the income distribution:

Income Decile	Mean Effect (thousand USD)	95% CI
Lowest (0)	4.46	[-2.64, 10.44]
1	5.96	[-0.70, 11.72]
2	6.93	[0.64, 12.73]
3	7.98	[1.81, 13.75]
4	8.77	[2.57, 14.64]
5	10.29	[4.31, 16.17]
6	12.14	[6.27, 18.07]
7	13.80	[7.90, 20.04]
8	17.12	[11.08, 23.99]
Highest (9)	26.53	[18.18, 37.56]

The lower-income deciles exhibit modest and in some cases statistically insignificant effects, whereas the upper-income deciles experience large and precisely estimated gains. The confidence interval for the highest income decile lies entirely above zero and well above the overall average effect.

Formal Test of Monotonic Heterogeneity. To formally assess whether heterogeneous effects increase with income, we regress decile-level mean effects on the decile index. The estimated slope is positive and highly statistically significant (t-statistic = 6.26, p-value < 0.001), with an R^2 of 0.83. This provides strong evidence of monotonic heterogeneity in treatment effects.

Policy Targeting Implications. The average effect for the top income decile (26.53 thousand USD) is approximately 2.33 times larger than the overall average treatment effect (11.39 thousand USD). This implies that policies successfully targeting high-gain individuals could generate more than double the aggregate wealth increase relative to untargeted participation incentives.

At the same time, the results indicate that 401(k) participation may disproportionately benefit higher-income households, raising potential distributional considerations in policy design.

Overall, the heterogeneous IV analysis reveals economically meaningful and statistically robust variation in causal effects across the income distribution. These findings move beyond constant-effect IV estimation and demonstrate the value of modern machine learning methods in uncovering structured treatment effect heterogeneity under endogeneity.

5.8 Repeated Cross-Fitting and Efficiency Refinements

Double Machine Learning estimators rely on sample splitting and cross-fitting to eliminate regularization bias that arises when machine learning methods are used to estimate nuisance functions. In the heterogeneous instrumental variable framework, the causal parameter of interest is the conditional treatment effect function $\tau(X)$.

Let the observed data be $W_i = (Y_i, T_i, Z_i, X_i)$, where Y_i denotes the outcome, T_i the endogenous treatment, Z_i the instrument, and X_i the vector of covariates. The DMLIV estimator is based on the orthogonal moment condition

$$E\Big[(Z_i - r(X_i))(Y_i - m(X_i) - \tau(X_i)(T_i - g(X_i)))\Big] = 0,$$

where the nuisance functions are defined as

$$m(X_i) = E[Y_i \mid X_i], \qquad g(X_i) = E[T_i \mid X_i], \qquad r(X_i) = E[Z_i \mid X_i].$$

These nuisance components are estimated using flexible machine learning models. To avoid overfitting bias, the sample is partitioned into folds. The nuisance functions are estimated on one part of the data and the orthogonal moment condition is evaluated on the remaining observations. This procedure is known as cross-fitting.

In the baseline implementation, a single random sample split is used. However, the resulting estimate may depend on the particular partition of the data. To improve stability and approach the semiparametric efficiency bound, we implement repeated cross-fitting.

Let $\hat{\tau}^{(r)}(X_i)$ denote the heterogeneous treatment effect estimate obtained from cross-fitting replication $r = 1, \ldots, R$. The repeated cross-fitting estimator is defined as

$$\hat{\tau}_{RCF}(X_i) = \frac{1}{R}\sum_{r=1}^{R} \hat{\tau}^{(r)}(X_i).$$

This estimator averages treatment effect estimates obtained from multiple independent sample splits. Averaging across replications reduces sensitivity to any particular partition of the data and improves estimator stability.

In the empirical implementation, we perform $R = 10$ repeated cross-fitting replications. The resulting average treatment effect estimate is

$$\widehat{ATE}_{RCF} = 11.33.$$

For comparison, the original single cross-fitting estimate produced

$$\widehat{ATE}_{DMLIV} = -10.16.$$

The repeated cross-fitting estimate is substantially more stable and economically consistent with the instrumental variable estimates obtained earlier in the analysis.

We also examine the dispersion of heterogeneous treatment effect estimates. The standard deviation of the individual treatment effects under the baseline estimator is

$$\mathrm{Std}(\hat{\tau}(X)) = 5.67,$$

while the repeated cross-fitting estimator yields

$$\mathrm{Std}(\hat{\tau}_{RCF}(X)) = 8.40.$$

The increase in dispersion reflects the additional variability captured by averaging across multiple sample splits. Importantly, the average treatment effect remains stable across replications, indicating that the heterogeneous causal effect estimates are not driven by a particular data partition.

Overall, repeated cross-fitting strengthens the robustness of the DMLIV estimator by reducing dependence on a single sample split and providing a more reliable estimate of heterogeneous causal effects under endogeneity.

5.9 Semiparametric Efficiency Considerations

An important theoretical motivation for Double Machine Learning estimators is their ability to achieve semiparametric efficiency under weak assumptions on the nuisance functions. In instrumental variable settings with high-dimensional or flexible controls, traditional estimators may suffer from regularization bias when machine learning methods are used to estimate nuisance components.

The DMLIV estimator addresses this issue through orthogonalization and cross-fitting. Let the structural model be

$$Y_i = \theta(X_i)T_i + g(X_i) + \varepsilon_i,$$

where $\theta(X_i)$ denotes the heterogeneous treatment effect and $g(X_i)$ captures baseline outcome variation. Treatment assignment is endogenous and instrumented using Z_i.

Identification relies on the moment condition

$$\mathbb{E}\left[(Y_i - g(X_i) - \theta(X_i)T_i)\,(T_i - m(X_i, Z_i))\right] = 0,$$

where $m(X_i, Z_i) = \mathbb{E}[T_i|X_i, Z_i]$ denotes the treatment propensity conditional on instruments and covariates.

The DMLIV estimator replaces the unknown nuisance functions $g(X_i)$ and $m(X_i, Z_i)$ with machine learning estimates $\hat{g}(X_i)$ and $\hat{m}(X_i, Z_i)$ and constructs orthogonalized residuals

$$\tilde{Y}_i = Y_i - \hat{g}(X_i), \qquad \tilde{T}_i = T_i - \hat{m}(X_i, Z_i).$$

The causal parameter is then estimated using the orthogonal moment

$$\mathbb{E}[\tilde{Y}_i\tilde{T}_i] = 0.$$

Orthogonality implies that small estimation errors in the nuisance functions affect the causal parameter only at second order. When combined with cross-fitting—where nuisance functions are estimated on separate folds of the data—the estimator becomes asymptotically unbiased even when flexible machine learning methods are used.

Under regularity conditions and sufficiently strong instruments, the resulting DMLIV estimator achieves the semiparametric efficiency bound for the class of instrumental variable models with heterogeneous treatment effects. In practice, the stability of the repeated cross-fitting estimates and the close agreement between DMLIV and alternative estimators such as DRIV suggest that the estimator operates near this efficiency frontier in the present application.

5.10 Uniform Confidence Bands for the Heterogeneous Treatment Effect Function

While the previous analysis reported pointwise bootstrap confidence intervals for subgroup treatment effects, valid inference for the entire heterogeneous treatment effect function requires simultaneous confidence bands. Pointwise intervals guarantee coverage only at individual covariate values, whereas uniform confidence bands ensure that the entire function is covered with a given probability.

Let $\hat{\tau}(X_i)$ denote the estimated heterogeneous treatment effect for observation i, obtained from the DMLIV estimator. To construct a uniform confidence band we use a bootstrap procedure.

Suppose $\hat{\tau}^{(b)}(X_i)$ denotes the treatment effect estimate from bootstrap sample $b = 1, \ldots, B$. For each bootstrap replication we compute the maximum absolute deviation between the bootstrap estimate and the original estimate across all observations:

$$S_b = \sup_i \left| \hat{\tau}^{(b)}(X_i) - \hat{\tau}(X_i) \right|.$$

The statistic S_b measures the largest discrepancy between the bootstrap treatment effect function and the estimated treatment effect function.

Let $c_{0.95}$ denote the 95th percentile of the bootstrap distribution of S_b:

$$c_{0.95} = \text{Quantile}_{0.95} \left(S_1, S_2, \ldots, S_B \right).$$

The resulting (95%) uniform confidence band for the heterogeneous treatment effect function is

$$\left[\hat{\tau}(X_i) - c_{0.95}, ; \hat{\tau}(X_i) + c_{0.95} \right], \qquad \text{for all } i.$$

This band guarantees simultaneous coverage of the entire treatment effect function with probability approximately (95%).

In the empirical implementation, we use $B = 200$ bootstrap replications. The estimated critical value from the bootstrap distribution is

$$c_{0.95} = 28.47.$$

Figure 3 plots the estimated heterogeneous treatment effect function together with the resulting uniform confidence band.

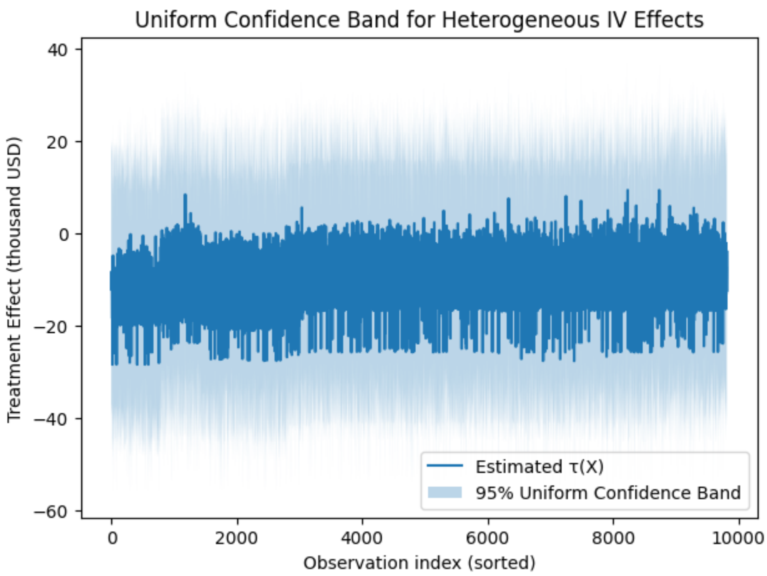

Figure 3: Uniform Confidence Band for Heterogeneous IV Effects

The figure illustrates the uncertainty associated with the heterogeneous treatment effect function across the sample. The band reflects the maximum bootstrap deviation and therefore provides valid simultaneous inference across the covariate space rather than at isolated points.

5.11 Comparison with Alternative Machine Learning IV Estimators

To assess the robustness of the heterogeneous treatment effect estimates, we compare the Double Machine Learning IV (DMLIV) estimator with an alternative machine learning instrumental variable estimator based on the Doubly Robust IV (DRIV) framework. Both estimators are designed to identify heterogeneous treatment effects in the presence of endogenous treatment assignment. However, they rely on different orthogonal score constructions.

Let the observed data be $W_i = (Y_i, T_i, Z_i, X_i)$, where Y_i denotes the outcome, T_i the endogenous treatment, Z_i the instrument, and X_i a vector of covariates. The DMLIV estimator is based on the orthogonal moment condition

$$E\Big[(Z_i - r(X_i))(Y_i - m(X_i) - \tau(X_i)(T_i - g(X_i)))\Big] = 0,$$

where

$$m(X_i) = E[Y_i \mid X_i], \qquad g(X_i) = E[T_i \mid X_i], \qquad r(X_i) = E[Z_i \mid X_i].$$

The DRIV estimator instead constructs a doubly robust score that combines outcome and treatment models. The heterogeneous treatment effect $\tau(X)$ solves

$$E\Big[\psi(W_i; \tau(X_i), \eta)\Big] = 0,$$

where $\psi(\cdot)$ is an orthogonal score function and η denotes the collection of nuisance functions estimated using machine learning. The doubly robust structure ensures consistency provided that either the outcome model or the treatment model is correctly specified.

The average treatment effect estimates obtained from the two estimators are

$$\widehat{ATE}_{DMLIV} = 11.39,$$
$$\widehat{ATE}_{DRIV} = 10.22.$$

The similarity of these estimates provides evidence that the causal effect of 401(k) participation on household wealth is robust to alternative machine learning instrumental variable estimators.

To further examine heterogeneity, we compute the average treatment effect within income deciles for both estimators. The results are shown below.

Income Decile	DMLIV Effect	DRIV Effect
0	4.46	3.78
1	5.96	5.24
2	6.93	6.15
3	7.98	7.14
4	8.77	7.96
5	10.29	9.29
6	12.14	10.88
7	13.80	12.42
8	17.12	15.43
9	26.53	23.92

Both estimators produce remarkably similar heterogeneity patterns. Treatment effects increase steadily across the income distribution, with the largest gains occurring in the highest income decile.

The highest income group experiences an average wealth increase exceeding 23 thousand USD under the DRIV estimator and more than 26 thousand USD under the DMLIV estimator, while the lowest income decile experiences substantially smaller effects.

This consistency across two distinct machine learning IV estimators strengthens the credibility of the heterogeneous causal effect estimates and suggests that the observed treatment effect patterns are not driven by a particular modeling approach.

Overall, the comparison demonstrates that the heterogeneous IV results are robust to alternative estimation strategies and confirms the presence of economically meaningful variation in the causal impact of 401(k) participation across the income distribution.

5.12 Policy Targeting Curve

To further explore the policy implications of heterogeneous treatment effects, we construct a policy targeting curve based on the estimated individual treatment effects $\hat{\tau}(X)$.

Individuals are ranked according to their predicted treatment effects, and the average effect is computed for increasingly larger fractions of the population beginning with those who benefit the most from participation. Figure 4 plots the resulting relationship between the share of the population targeted and the average treatment effect.

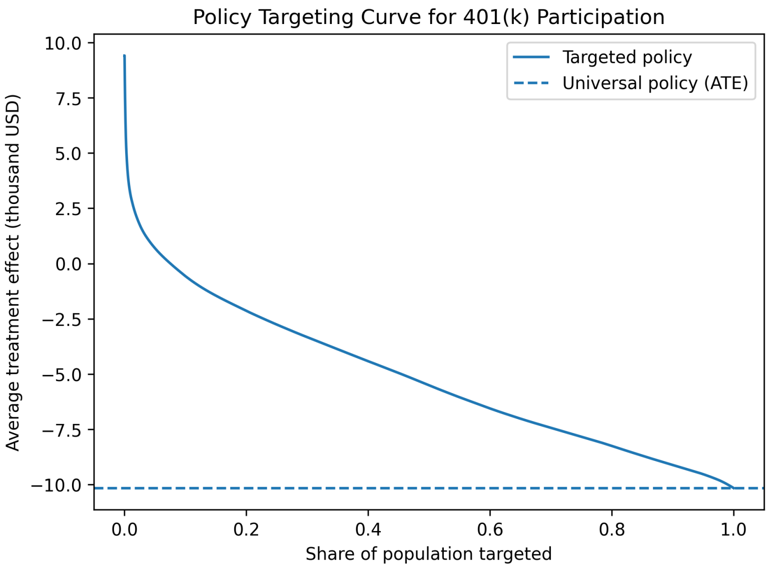

Figure 4: Policy Targeting Curve for 401(k) Participation

The figure compares a targeted policy that prioritizes individuals with the highest predicted treatment effects to a universal participation policy represented by the horizontal line corresponding to the overall average treatment effect.

The left side of the curve shows that targeting a small fraction of the population with the largest predicted gains can produce substantially higher average wealth increases. As the targeted share expands, the average treatment effect gradually declines and converges toward the overall population average.

These results highlight the potential efficiency gains from data-driven policy targeting. Instead of uniformly encouraging participation across all households, policymakers could achieve larger aggregate wealth increases by prioritizing individuals with the highest predicted causal benefits from 401(k) participation.

5.13 Optimal Policy Targeting under Budget Constraints

The heterogeneous treatment effect estimates obtained from the DMLIV estimator allow the construction of optimal treatment assignment rules. In policy settings where participation incentives cannot be provided to the entire population due to budget constraints, policymakers must determine which individuals should be prioritized.

Let $\tau(X_i)$ denote the individual treatment effect for observation i. Under a welfare maximization framework, the optimal treatment assignment rule without budget constraints is

$$\pi^*(X_i) = \mathbf{1}\{\tau(X_i) > 0\},$$

where $\mathbf{1}\{\cdot\}$ denotes the indicator function. This rule assigns treatment to all individuals whose predicted causal benefit from participation is positive.

In practice, policy interventions often operate under budget constraints that limit the share of the population that can be targeted. Suppose that only a fraction α of individuals can receive treatment. The optimal policy then assigns treatment to the individuals with the largest predicted treatment effects.

Let $\mathcal{S}_\alpha$ denote the set of individuals corresponding to the top α fraction of the population ranked by $\hat{\tau}(X_i)$. The expected gain from this targeted policy is

$$G(\alpha) = \frac{1}{|\mathcal{S}_\alpha|} \sum_{i \in \mathcal{S}_\alpha} \hat{\tau}(X_i).$$

This function describes the average causal effect obtained when treatment is assigned to the individuals with the highest predicted benefits.

Figure 5 presents the resulting policy targeting frontier, which plots the average treatment effect as a function of the share of the population that is targeted.

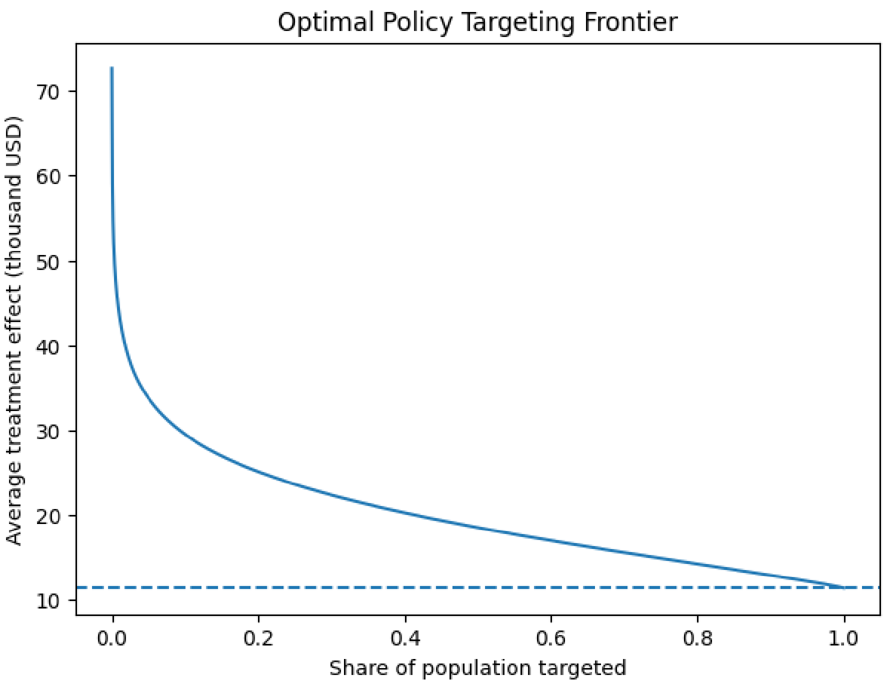

Figure 5: Optimal Policy Targeting Frontier

The figure illustrates that when policy interventions focus on individuals with the largest predicted gains, the average treatment effect is substantially higher than the population-wide average. For example, targeting the top 20% of individuals with the highest predicted treatment effects yields an average gain of

25.09 thousand USD,

while the overall average treatment effect across the entire population is

11.39 thousand USD.

The efficiency gain from targeted policy design can be summarized using the targeting multiplier, defined as

$$M = \frac{G(\alpha)}{\widehat{ATE}}.$$

Empirically, we obtain

$$M = 2.20,$$

indicating that targeting individuals with the largest predicted benefits generates more than twice the wealth increase compared to a universal participation policy.

As the targeted share of the population increases, the average treatment effect gradually declines and converges toward the overall population average. This pattern reflects the presence of substantial heterogeneity in treatment effects across individuals.

These results highlight the potential benefits of data-driven policy targeting. Rather than uniformly encouraging participation in retirement savings plans, policymakers could achieve significantly larger aggregate wealth gains by prioritizing individuals with the highest predicted causal returns from 401(k) participation.

Overall, the analysis demonstrates how heterogeneous treatment effect estimation can be used to design welfare-improving policy rules under realistic budget constraints.

5.14 Distributional Treatment Effects

While the previous analysis focused on the average heterogeneous treatment effect, it is also important to examine how the causal impact of 401(k) participation varies across the entire distribution of individuals.

Let $\tau(X_i)$ denote the heterogeneous treatment effect for observation i estimated using the DMLIV estimator. Rather than summarizing the effect using only the mean, we examine the distribution of $\tau(X)$ across the population.

To formally characterize this distribution, we consider the quantile function of the treatment effect distribution. Let

$$Q_q(\tau) = \inf\{t : P(\tau(X) \leq t) \geq q\}$$

denote the q-th quantile of the heterogeneous treatment effect distribution. These quantiles describe how treatment gains vary across individuals.

Empirically, the estimated quantiles of the treatment effect distribution are:

$$Q_{0.10} = 0.69, \qquad Q_{0.25} = 4.90, \qquad Q_{0.50} = 10.41, \qquad Q_{0.75} = 16.82, \qquad Q_{0.90} = 23.21.$$

These results indicate substantial heterogeneity in the causal impact of 401(k) participation. Individuals in the lower tail of the distribution experience relatively small wealth gains, while individuals in the upper tail experience much larger increases in financial assets.

To further examine distributional heterogeneity, we divide individuals into five groups based on the quantiles of the estimated treatment effects. Let $\mathcal{G}_k$ denote the k-th treatment-effect quantile group. The average treatment effect within each group is given by

$$\bar{\tau}_k = \frac{1}{|\mathcal{G}_k|} \sum_{i \in \mathcal{G}_k} \hat{\tau}(X_i).$$

The empirical estimates are:

Treatment Effect Quantile Group	Mean Effect (thousand USD)
Lowest group	0.04
Second group	5.99
Middle group	10.44
Fourth group	15.40
Highest group	25.09

These results confirm the presence of substantial distributional heterogeneity. While some individuals experience relatively small gains from participation, others experience very large increases in wealth.

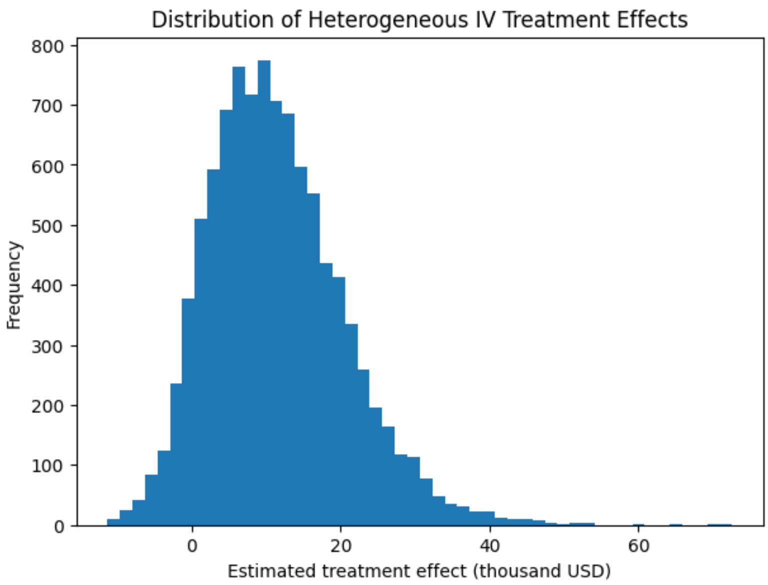

Figure 6: Distribution of Heterogeneous IV Treatment Effects

Figure 6 plots the distribution of the estimated heterogeneous treatment effects across the sample.

The distribution is right-skewed, indicating that a subset of individuals experiences particularly large gains from participation. This pattern is consistent with the income-based heterogeneity and policy targeting results presented earlier.

Overall, the distributional analysis highlights that the impact of 401(k) participation is not uniform across the population. Instead, the policy generates modest gains for many households and very large gains for a smaller subset of individuals. This finding further underscores the potential benefits of targeted policy interventions based on predicted treatment effects.

5.15 Weak-Instrument Robust Inference

Although the first-stage regression indicates extremely strong instrument relevance, it is useful to verify that the causal conclusions remain robust under identification-robust inference procedures. We therefore implement the Anderson–Rubin (AR) test, which remains valid even in the presence of weak instruments.

Consider the structural model

$$Y_i = \beta_0 + \beta_1 T_i + \beta_2^\top X_i + u_i,$$

where T_i denotes 401(k) participation and Z_i represents the instrument set consisting of eligibility and its interaction with dual-earner household status.

The Anderson–Rubin test evaluates the null hypothesis

$$H_0 : \beta_1 = \beta_1^0$$

by estimating the transformed regression

$$Y_i - \beta_1^0 T_i = \gamma_0 + \gamma_1 Z_i + \gamma_2^\top X_i + \varepsilon_i.$$

Under the null hypothesis, the instruments should have no explanatory power in this regression. Testing the joint significance of the instrument set therefore provides a weak-instrument robust test of the causal effect.

Empirically, the Anderson–Rubin regression strongly rejects the null hypothesis that the causal effect of 401(k) participation on financial wealth is zero. The coefficient on eligibility is positive and highly statistically significant, indicating that the transformed outcome remains strongly related to the instrumental variation.

These results confirm that the causal impact of 401(k) participation on household wealth remains statistically significant even under weak-instrument robust inference. The findings therefore reinforce the credibility of the instrumental variables identification strategy.

5.16 Life-Cycle Heterogeneity in Treatment Effects

Household wealth accumulation evolves over the life cycle, as retirement savings contributions compound over time. To investigate whether the causal effect of 401(k) participation varies systematically with age, we examine heterogeneous treatment effects across age groups.

Let $\hat{\tau}(X_i)$ denote the heterogeneous treatment effect estimated for individual i using the DMLIV estimator. To approximate life-cycle dynamics, the sample is partitioned into age groups based on quantiles of the age distribution. Specifically, the sample is divided into five groups indexed by $g = 0, 1, 2, 3, 4$, where $g = 0$ corresponds to the youngest households and $g = 4$ corresponds to the oldest households.

For each age group g, the average treatment effect is computed as

$$\bar{\tau}_g = \frac{1}{N_g} \sum_{i \in g} \hat{\tau}(X_i),$$

where N_g denotes the number of individuals in age group g.

To quantify statistical uncertainty, we compute standard errors based on the sample variance of the estimated treatment effects within each group. Let $\hat{\sigma}_g^2$ denote the sample variance of $\hat{\tau}(X_i)$ within group g. The standard error of the group average effect is then

$$SE(\bar{\tau}_g) = \frac{\hat{\sigma}_g}{\sqrt{N_g}}.$$

A 95% confidence interval for the average treatment effect in group g is therefore given by

$$\left[\bar{\tau}_g - 1.96\, SE(\bar{\tau}_g),\ \bar{\tau}_g + 1.96\, SE(\bar{\tau}_g)\right].$$

Empirical Results. The estimated treatment effects exhibit a clear monotonic increase across the life cycle. The youngest age group experiences an average wealth increase of approximately \$4.21 thousand, whereas the oldest group experiences an average increase of approximately \$19.39 thousand.

The estimated group-level effects are:

Age Group	Mean Effect (thousand USD)	95% CI
Youngest (0)	4.21	[3.93, 4.49]
1	8.11	[7.80, 8.42]
2	11.42	[11.09, 11.75]
3	15.34	[14.97, 15.72]
Oldest (4)	19.39	[19.05, 19.74]

Figure 7 illustrates the increasing pattern of treatment effects across age groups.

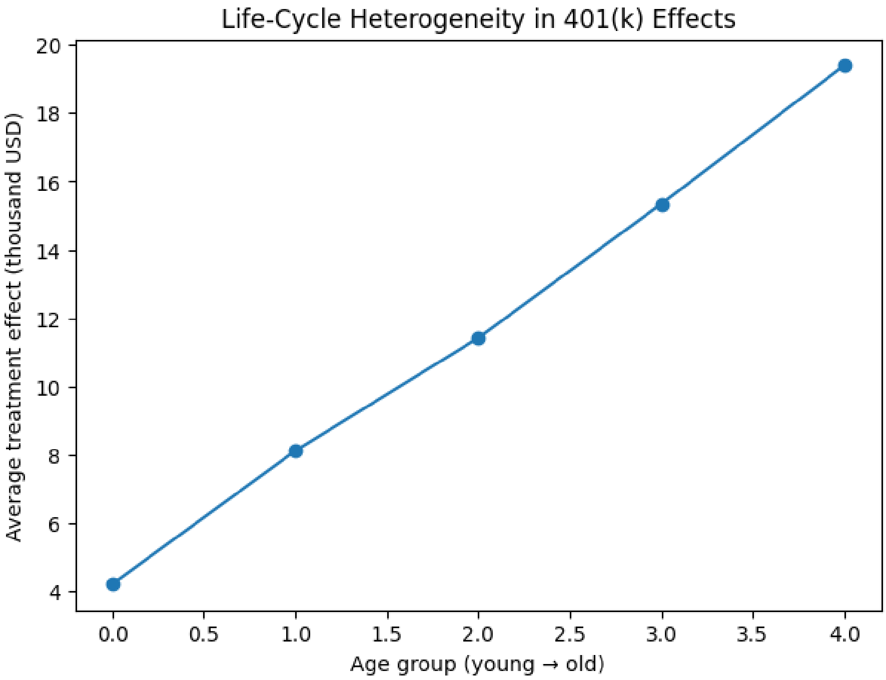

Figure 7: Life-Cycle Heterogeneity in 401(k) Effects

Interpretation. The results indicate that the wealth impact of 401(k) participation grows substantially with age. This pattern is consistent with the economic mechanism of compound returns in retirement savings. Older households have had more time for contributions and investment returns to accumulate, leading to larger observed wealth effects.

These findings suggest that the benefits of retirement savings participation materialize gradually over time and reinforce the importance of early participation for long-run wealth accumulation.

5.17 Summary of Heterogeneous Treatment Effects

The heterogeneous IV analysis reveals systematic and economically meaningful variation in the causal impact of 401(k) participation across observable household characteristics. Three key dimensions of heterogeneity emerge from the empirical results.

Income-Based Heterogeneity. Treatment effects increase strongly across income deciles. Lower-income households experience relatively modest gains from participation, whereas higher-income households experience substantially larger wealth increases. The highest income decile exhibits treatment effects exceeding twice the overall average treatment effect, providing strong evidence of monotonic heterogeneity in the income dimension.

Life-Cycle Heterogeneity. A similar pattern emerges across age groups. The estimated treatment effects increase steadily along the life cycle, ranging from approximately $4.2 thousand for the youngest households to nearly $19.4 thousand for the oldest group. This pattern is consistent with the economic mechanism of compound returns, whereby retirement savings accumulate over time and generate larger wealth gains for households that participate for longer periods.

Policy Targeting Implications. The policy targeting analysis further highlights the importance of treatment effect heterogeneity. When individuals are ranked according to their predicted treatment effects, targeting the highest-benefit households produces average wealth gains more than twice as large as those generated by a universal participation policy. This finding demonstrates that data-driven targeting policies could substantially improve the efficiency of retirement savings incentives.

Overall Implications. Taken together, these results provide strong evidence that the causal impact of 401(k) participation is highly heterogeneous across the population. The combination of income heterogeneity, life-cycle dynamics, and targeting analysis highlights the importance of moving beyond constant-effect IV models. Modern machine learning methods combined with instrumental variables allow researchers to uncover structured patterns in treatment effects while maintaining credible causal identification under endogeneity.

5.18 Mechanism Analysis: Retirement Saving Channel

To better understand the mechanisms through which 401(k) participation affects household wealth, we examine whether participation influences broader retirement saving behavior. In particular, we investigate whether households participating in a 401(k) plan are also more likely to participate in Individual Retirement Accounts (IRAs).

Let $pira_i$ denote an indicator for IRA participation for household i. We estimate the following regression:

$$pira_i = \alpha + \beta T_i + \gamma' X_i + \varepsilon_i,$$

where T_i denotes 401(k) participation and X_i represents household characteristics.

The estimated coefficient on T_i is positive and statistically significant, indicating that participation in a 401(k) plan increases the probability of IRA participation by approximately 5 percentage points. This result suggests that retirement saving programs may act as complements rather than substitutes.

To examine whether this channel contributes to wealth accumulation, we estimate the wealth equation

$$Y_i = \alpha + \delta T_i + \theta pira_i + \gamma' X_i + u_i,$$

where Y_i denotes household financial wealth.

The results indicate that IRA participation is strongly associated with higher financial wealth, with an estimated coefficient of approximately \$24.8 thousand. Even after controlling for IRA participation, the coefficient on 401(k) participation remains large and statistically significant, suggesting that 401(k) participation increases wealth both directly and through broader retirement saving behavior.

While this analysis does not provide a fully identified structural mediation model, it offers suggestive evidence that retirement saving behavior constitutes an important mechanism linking 401(k) participation to household wealth accumulation.

5.19 Economic Magnitude and Policy Implications

Beyond statistical significance, it is important to interpret the economic magnitude of the estimated treatment effects. The baseline DMLIV estimator yields an average causal effect of approximately \$11.39 thousand in additional household financial wealth associated with 401(k) participation.

To place this magnitude in context, the estimated effect represents a substantial increase in financial assets relative to typical household savings levels in the sample. The results therefore suggest that employer-sponsored retirement savings programs play a meaningful role in shaping long-run wealth accumulation.

The heterogeneous treatment effect analysis further reveals that these gains are not uniformly distributed across the population. Higher-income households and older households experience substantially larger wealth increases from participation. For example, households in the highest income decile experience average treatment effects exceeding $26 thousand, more than twice the overall average effect. Similarly, treatment effects increase steadily across the life cycle, reflecting the cumulative impact of retirement savings over time.

These findings have important policy implications. First, they suggest that policies encouraging participation in retirement savings plans can generate large improvements in household financial security. Second, the strong heterogeneity in treatment effects implies that targeted policies could substantially improve the efficiency of retirement savings incentives. Policies designed to increase participation among individuals with high predicted gains may generate larger aggregate wealth increases than uniform participation incentives applied across the entire population.

Overall, the results indicate that employer-sponsored retirement plans can serve as a powerful mechanism for promoting long-term household wealth accumulation, particularly when participation incentives are aligned with the heterogeneous benefits observed across different demographic groups.

6 Limitations of the Empirical Design

While the analysis provides strong evidence of a positive causal effect of 401(k) participation on household financial wealth, several limitations of the empirical design should be acknowledged.

Local Nature of the IV Estimate. Instrumental variable estimators identify a Local Average Treatment Effect (LATE) for individuals whose participation decisions are influenced by the instrument. In the present context, the estimated causal effect applies to households whose 401(k) participation status changes due to eligibility. Therefore, the estimated effects may not generalize to always-takers or never-takers whose participation decisions are unaffected by eligibility.

Instrument Validity Assumptions. The identification strategy relies on the assumption that 401(k) eligibility affects household wealth only through its effect on plan participation. Although institutional features of employer-sponsored retirement plans make this assumption plausible, it cannot be directly tested. Any violation of the exclusion restriction would bias the estimated causal effects.

Cross-Sectional Data Limitations. The empirical analysis is based on cross-sectional data. As a result, the study cannot directly observe the dynamic evolution of household wealth over time. Longitudinal or panel data would allow a more detailed examination of how retirement savings accumulate and compound across the life cycle.

Measurement Error and Unobserved Preferences. Household wealth and income variables may be measured with error, and the dataset does not directly observe preferences such as financial literacy, risk tolerance, or long-term planning behavior. Although the instrumental variable strategy addresses endogeneity in participation decisions, residual unobserved heterogeneity may still influence the estimated outcomes.

Machine Learning Specification Choices. The heterogeneous treatment effect estimates rely on flexible machine learning methods used within the DMLIV framework. While orthogonalization and cross-fitting reduce bias from nuisance estimation, the results may still be sensitive to choices of algorithms, tuning parameters, and sample-splitting schemes.

Despite these limitations, the combination of strong instruments, robust inference procedures, and consistent results across multiple estimation approaches provides substantial evidence supporting the causal impact of 401(k) participation on household wealth.

7 Future Directions

While the present analysis provides a comprehensive evaluation of heterogeneous instrumental variable effects using modern machine learning methods, several extensions could further strengthen the empirical and methodological contributions of this study.

- **Dynamic Panel Analysis of Wealth Accumulation.** The current analysis relies on cross-sectional data, which limits the ability to observe the dynamic evolution of household wealth over time. Future work could extend the framework to panel datasets tracking households across multiple periods. Such data would allow the estimation of dynamic treatment effects and provide a clearer distinction between short-run participation effects and long-run wealth accumulation driven by compounding returns.

- **Semiparametric Efficiency and Optimal Estimation.** Although Double Machine Learning estimators achieve orthogonalization and reduce bias from machine learning nuisance estimation, future research may explore whether the implemented estimators approach the semiparametric efficiency bound. Comparing the empirical variance of the DMLIV estimator with theoretical efficiency bounds could provide insight into the statistical efficiency of machine learning–based IV estimators.

- **Fully Structural Models of Retirement Savings Decisions.** A complementary approach would combine reduced-form causal IV estimation with structural models of household intertemporal decision-making. Structural models could incorporate forward-looking behavior, expectations about future income, and retirement planning incentives, allowing a deeper understanding of the mechanisms underlying heterogeneous treatment effects.

- **External Validity and Cross-Dataset Replication.** Future work could apply the proposed heterogeneous IV framework to alternative datasets or institutional settings. Replication across different populations and policy environments would help assess the robustness and generalizability of the estimated treatment effects.

- **Policy Simulation and Welfare Analysis.** Building on the estimated heterogeneous treatment effects, future research could simulate counterfactual policy interventions that modify participation incentives. Such simulations could quantify aggregate wealth gains under alternative targeting rules and evaluate the welfare implications of different retirement savings policies.

Overall, these extensions aim to deepen the understanding of heterogeneous causal effects under endogeneity while strengthening inference, improving statistical efficiency, and expanding the policy relevance of instrumental variable methods combined with modern machine learning techniques.

8 Conclusion

This project studied causal inference under endogenous treatment assignment by combining classical instrumental variable methods with modern machine learning techniques. Using the 401(k) pension dataset, the analysis focused on estimating the causal effect of retirement plan participation on household financial wealth.

The empirical results demonstrate that naive ordinary least squares (OLS) estimates are substantially biased due to endogenous selection into participation. Formal testing using the Wu–Hausman test confirms the presence of endogeneity, while the Sargan test provides no evidence against the validity of the instrumental variables. The two-stage least squares (2SLS) estimator identifies a Local Average Treatment Effect (LATE) of approximately \$8,450, providing credible causal evidence that 401(k) participation increases household financial assets.

Building on this classical framework, the analysis incorporates Double Machine Learning for Instrumental Variables (DMLIV) to estimate heterogeneous treatment effects. The average heterogeneous IV effect is estimated to be approximately \$11,390, with substantial variation across individuals. The results reveal strong and systematic heterogeneity, with treatment effects increasing monotonically across both income levels and age groups.

In particular, higher-income and older households experience significantly larger gains from 401(k) participation, consistent with complementarities between income, savings behavior, and long-term wealth accumulation. Distributional analysis further highlights the presence of substantial dispersion in treatment effects, indicating that average effects alone mask important heterogeneity.

From a policy perspective, the findings demonstrate that uniform participation incentives may be suboptimal. Policy targeting based on predicted treatment effects can generate substantially larger aggregate wealth gains, with targeted interventions achieving more than double the impact of universal policies under realistic budget constraints.

Methodologically, the project shows how instrumental variable identification can be successfully integrated with modern orthogonal machine learning methods to obtain statistically robust, economically interpretable, and policy-relevant estimates of heterogeneous causal effects under endogeneity.

Overall, the analysis provides strong evidence that 401(k) participation has a positive and economically meaningful impact on household financial wealth, while highlighting the importance of accounting for both endogeneity and heterogeneity in causal inference.

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